A CLINICAL DECISION SUPPORT SYSTEM

AI-powered pediatric pneumonia detection.

Saving children's lives — one chest X-ray at a time. Built for under-resourced hospitals across Algeria.

0.981	95.01%	<100ms	2.5M+
AUC-ROC	SENSITIVITY	PREDICTION SPEED	CHILD DEATHS / YR

PRESENTED BY DR. ABDERRAHMANE KHIAT · UNIVERSITY OF SAIDA, ALGERIA2026

A diagnostic gap children cannot cannot afford.

Pneumonia remains the leading infectious cause of pediatric mortality — concentrated in the countries least equipped to detect it.

2.5M

CHILD DEATHS / YEAR

Leading infectious cause of child mortality (WHO 2023). The vast majority occur in low-income countries.

1 : 500K

PEDIATRIC RADIOLOGIST RATIO

Pediatric radiology is critically scarce in Algeria. Children wait 6–24 hours for rural diagnosis.

40%

TRAVEL > 50 KM FOR CARE

Nearly half of pediatric cases must travel long distances. No local access to X-ray facilities.

20–30%

MISSED DIAGNOSES

Junior clinicians miss pneumonia at this rate. Delayed treatment in under-5s can be rapidly fatal.

200–300

ER CASES / DAY

Algerian pediatric emergency rooms are overwhelmed daily, with limited specialist support available.

70–85%

INTER-RATER AGREEMENT

Even experts disagree on borderline cases. An AI baseline can reduce variability and clinical errors.

Three customer segments across health and health and tech.

Designed to integrate at every layer of pediatric care delivery — from frontline hospitals to digital health partners and academic research.

SEGMENT 01

Hospitals & Pediatric Emergency

- →Public hospitals and clinics with pediatric units
- →Facilities with 200–300 respiratory cases / day
- →Centers with digital X-ray and patient intake forms
- →Priority: CHU and wilaya hospitals

SEGMENT 02

Health-Tech & Telehealth

- →Telemedicine platforms serving rural Algeria
- →Digital health startups building remote diagnostics
- →Medical device companies seeking AI-enhanced CXR
- →HIS vendors needing HL7 / FHIR integration

SEGMENT 03

Medical Training & Research

- →Medical schools using AI for radiology training
- →Research hospitals validating AI-assisted diagnosis
- →Ministry of Health digital health programs
- →International NGOs (MSF, WHO) field tools

A multidisciplinary team.

AI research, clinical medicine, and business strategy — under the supervision of Dr. Abderrahmane Khiat.

FOUNDER & SUPERVISOR

Dr. Abderrahmane Khiat

PhD Computer Science · AI systems & digital health. Former Project Lead at Fraunhofer IAIS, IT Project Manager at Hyundai AutoEver.

MEDICAL ADVISOR

Dr. Aimer Mohammed D.E.

Clinical validation of AI outputs. Confirms diagnostic thresholds and ensures medical safety standards across the project.

PM & ML LEAD

Kassouar Fatima

CSV preprocessing · 7-model training · manager · .

DL ENGINEER

Labani Nabila N.E.

CNN training · Grad-CAM · threshold optimisation · external validation.

DATA ENGINEER

Bouhmidi Amina Maroua

X-ray preprocessing · stratified split · augmentation · alignment.

BUSINESS ANALYST

Miloudi Meroua Amria

Market analysis · pricing · subscription strategy · partnerships.

Multimodal AI for pediatric pneumonia.

Two independent diagnostic branches — chest X-ray and clinical symptoms — combined into a single confidence-aware decision.

BRANCH A

CNN image branch

DenseNet121 on 5,856 pediatric X-rays

AUC 0.981 · Sensitivity 95.01%

Grad-CAM heatmaps for explainability

BRANCH B

Clinical CSV branch

Gradient Boosting on 127 pediatric records

AUC 0.996 · zero false positives

Key features: fatigue, fever, SpO2

FUSION

Multimodal late fusion

Combines clinical symptoms + X-ray evidence

Weighted by AUC performance per branch

One unified confidence score output

<100ms

Real-time prediction

Faster than any radiologist. Runs on standard hospital hardware — no GPU required.

3-tier safety

Pneumonia / Normal / Uncertain

Borderline cases auto-routed to expert review. Clinical safety is built in.

Grad-CAM

Visual explainability

Heatmaps overlay lung regions, building radiologist trust and supporting regulatory compliance.

Six steps from intake to diagnosis.

A clinical pipeline designed to slot into existing pediatric workflows — never replacing the physician, always supporting them.

01

Patient arrives

Nurse records symptoms and vitals. Patient fills clinical intake form. Radiographer captures the chest X-ray.

02

Parallel AI analysis

CNN analyses the X-ray image. Clinical model analyses symptoms. Both branches run simultaneously.

03

Confidence score

Each branch outputs a likelihood. Scores reflect strength of evidence and are ready for fusion.

04

Weighted fusion

Both branch scores combined. The better-performing model receives more weight, producing a unified score.

05

Three-tier decision

High score → Pneumonia. Low score → Normal. Borderline → routed to a radiologist for expert review.

06

Physician confirms

Doctor reviews AI suggestion. Grad-CAM highlights key regions. Physician makes the final diagnosis.

PIPELINE Sensors (oxygen, fever, chest pain, age) feed clinical data to AI · App + X-ray device process data in parallel · App suggests Pneumonia / Normal with Grad-CAM · System monitors physician confirmation rates to refine recommendations.

Three validated scientific hypotheses.

Built on peer-reviewed research and confirmed through bootstrap validation on the Algerian pediatric cohort.

H1 MULTIMODAL SUPERIORITY

Combining clinical symptoms with chest X-rays achieves higher sensitivity than either source alone. The two modalities provide complementary, independent diagnostic evidence.

H2 DENSENET121 ARCHITECTURE

Dense feature connectivity enables maximum reuse of learned visual patterns — optimal for detecting subtle lung consolidation in pediatric X-rays. Confirmed by Stanford CheXNet.

H3 FATIGUE AS DOMINANT PREDICTOR

Fatigue is a clinically validated dominant predictor of pneumonia in children. The Gradient Boosting model exploits this effectively even on a small clinical dataset (feature importance = 0.648).

SOURCES

Rajpurkar et al. — CheXNet, Stanford 2017 · Huang et al. — Multimodal Fusion Clinical AI, 2020 · WHO Pneumonia Fact Sheet 2023 · Dietterich — Ensemble Methods in ML, 2000. Bootstrap validation confirms the CSV model generalises reliably on n = 127.

Three real-world cases, one decision system. system.

CASE 01

Clear pneumonia

3 yrs · high fever · severe fatigue · cough · opacity on X-ray

Clinical HIGH confidence

X-ray HIGH confidence

Final **PNEUMON IA**

Pneumonia detected

Both branches agree. Grad-CAM highlights consolidation in the right lower lobe.

CASE 02

Clear normal

2 yrs · no fever · no cough · no fatigue · clear lung fields

Clinical LOW likelihood

X-ray LOW likelihood

Final **NORMAL**

No action required

Both branches agree — very low pneumonia likelihood. No treatment indicated.

CASE 03

Uncertain

4 yrs · mild fever · borderline X-ray · subtle haziness in left field

Clinical MODERATE

X-ray MODERATE

Final **UNCERTAIN**

Routed to radiologist

Borderline case. Flagged for immediate expert review — the safety feature that prevents misclassification.

PNEUMO-AI vs. traditional consultation.

Clear advantages across every dimension — speed, accuracy, accessibility, and explainability.

DIMENSION	TRADITIONAL CONSULTATION	PNEUMO-AI SYSTEM
Speed	5–20 min per X-ray reading	< 100 ms per prediction
Availability	1 radiologist per 500K children	Unlimited — available 24 / 7
Accuracy	20–30% junior miss rate	95%+ sensitivity after optimisation
Modality	Single: X-ray or symptoms	Dual: X-ray and symptoms
Confidence	No confidence score given	3-tier output + uncertainty routing
Scalability	Expert only — not scalable	Deployable on standard hospital PC
Explainability	Black-box diagnosis	Grad-CAM heatmap shows reasoning
EMR integration	No EMR integration	HL7 / FHIR integration planned

A reproducible, tested prototype.

Documented across 10 Jupyter notebooks — from raw preprocessing to final fused prediction.

CNN BRANCH · DENSENET121

- →DenseNet121 / ResNet50 / VGG16 on 5,856 X-rays
- →Corrected 70 / 15 / 15 stratified split
- →Threshold optimisation loop [0.01–0.99]
- →Grad-CAM heatmap generation across 6 cases
- →External validation: sensitivity 97.21%

CSV CLINICAL BRANCH

- →7 classifiers: GradBoost, RF, SVM, KNN, NB, LR, MLP
- →Pediatric filter: 710 → 127 patients (ages 1–16)
- →Bootstrap + LOOCV + Repeated K-Fold validation
- →Feature importance — fatigue = 64.8%
- →All outputs saved to GitHub + Google Drive

LATE FUSION INTEGRATION

- →AUC-weighted combination (DL + ML)
- →Three-tier output: Pneumonia / Normal / Uncertain
- →Confidence-aware routing to radiologist
- →Ready for paired evaluation (MIMIC-CXR)
- →Full pipeline: raw data → final prediction

10 JUPYTER NOTEBOOKS preprocessing · EDA · data loaders · CSV training · CNN training · evaluation · Grad-CAM · fusion · external validation · GitHub & Google Drive

DenseNet121 wins on every clinical criterion.

Systematic comparison across all candidate architectures, on both image and clinical branches.

BRANCH A — CNN MODELS (CHEST X-RAY)

MODEL	AUC	SENSITIVITY	SPECIFICITY	ACCURACY	PARAMS
DenseNet121 ★ Best	0.981	95.01%	92.44%	94.31%	8 M
ResNet50	0.880	95.48%	31.51%	78.16%	25 M
VGG16	0.964	95.32%	80.67%	91.35%	138 M

BRANCH B — CSV CLINICAL MODELS

MODEL	AUC	ACCURACY	PRECISION	RECALL	CV AUC
Gradient Boost ★ Best	0.996	93.75%	1.000	85.71%	0.9939 ± 0.009
Random Forest	0.978	93.75%	1.000	85.71%	0.9937 ± 0.007
Naive Bayes	0.984	87.50%	1.000	71.43%	0.9789 ± 0.015

Every clinical target met.

Validated on held-out test sets across both branches — and on an external dataset.

0.981	95.01%	92.44%	94.31%	97.21%
AUC-ROC (CNN)	SENSITIVITY (CNN)	SPECIFICITY (CNN)	ACCURACY (CNN)	EXTERNAL SENSITIVITY
Target met	Target met	Target met	Target met	Target met
0.996	1.000	93.75%	57.97m	All pass
AUC (CSV BRANCH)	PRECISION (CSV)	ACCURACY (CSV)	CNN TRAINING TIME	BOTH BRANCHES
Target > 0.95	Zero false positives	Target ≥ 85%	T4 GPU / Colab	Validated

A 90-day path to first hospital test.

From research artefact to validated clinical deployment in three months.

MONTH 01

Foundation

- Submit paper to IEEE / Springer pediatric AI journal
- Formal meeting with CHU Saida to propose pilot
- Apply for PhysioNet / MIMIC-CXR data access
- Draft IRB / ethics committee application

MONTH 02

Validation

- Begin clinical dataset expansion — target 500 real records
- Head-to-head comparison vs. attending radiologist (CHU)
- Register for Algerian Digital Health Innovation Fund (ANDS)
- First meeting with CNAS on AI-triage cost reduction

MONTH 03

Launch

- Deploy web interface demo for clinician feedback (50 users)
- Pediatric Medicine Days — Oran 2026
- Apply: WHO EMRO innovation grant + Fraunhofer collaboration
- Publish open-source Grad-CAM module on GitHub

OUR ASK Pilot partnership with CHU Saida · Ethics committee approval · ANDS / WHO grant access · Clinical advisor network.

BUSINESS MODEL

Business side

Market strategy · Business model · Partnerships · Go-to-market.

FRAME

KHIAT Startup

CURRENCY

DZD pricing only

STRATEGY

Free-first, then SaaS

Y3 TARGET

50–150 M DZD revenue

Six values: clinical precision meets predictable economics.

Three medical proofs delivered to physicians and patients — three institutional proofs delivered to healthcare organisations.

MEDICAL VALUE

Dual-pathway diagnostic precision

GBM classifier on 12 vital-sign parameters (fatigue importance 64.8%) combined with DenseNet121 CNN at 95% sensitivity, plus Grad-CAM visual explainability — for children aged 1–16.

RAG-powered decision support

Hybrid FAISS + BM25 retrieval queries patient records and a curated medical knowledge base in real time — risk identification, treatment-failure detection, and evidence-cited recommendations under 200 words.

Automated medical documentation

Patient records, treatment plans, monitoring tables, and AI-generated reports produced automatically — eliminating transcription overhead and enabling clean handover without manual documentation.

BUSINESS VALUE

Predictable cost efficiency

Storage costs are negligible (iBox 100 GB / 8,000 DZD/yr covers all scales). AI inference is the sole variable cost — controlled via caching and single-call generation, enabling 3×–5× cost reduction.

Scalable SaaS architecture

A single codebase serves cabinets (30 pts/day), clinics (100 pts/day), and hospitals (200 pts/day) with marginal incremental cost. File-based persistence deploys on any machine — no database server required.

Competitive data moat

The structured medical data lake (CSVs, X-rays, treatment PDFs) is a monetisable asset — sold to research labs and universities from Year 2, powering a specialised Algerian medical LLM from Year 3.

Algeria's healthcare digitalisation, 2024–2030. 2030.

A market accelerating under Plan Santé 2025 — with the pediatric segment structurally underserved.

48M
POPULATION

1,200+
PUBLIC HOSPITALS

5,000+
PRIVATE CLINICS

15+
MEDICAL UNIVERSITIES

MARKET SEGMENT	2024 VALUE (DZD)	CAGR 2024–2030	CONTEXT
Healthcare IT & EMR Systems	~40 Bn	~12% / yr	Digitalisation of public hospitals (Plan Santé 2025).
Clinical Decision Support (CDSS)	~12 Bn	~15% / yr	Growing adoption in CHUs and private clinics.
Pediatric & Emergency Market	~18 Bn	~10% / yr	Network of 1,200+ hospitals, 5,000+ private clinics.
Medical Research & Data	~5 Bn	~18% / yr	Rising demand for real clinical datasets for research.

Predictable economics across three deployment scales. scales.

Storage is effectively zero (iBox 100 GB · 8,000 DZD/yr covers every scale). AI inference is the only variable cost — and it scales linearly with patient volume.

CABINET

Individual physician

Daily volume	30 pts
Monthly volume	900 pts
Storage / month	4.5 GB
AI inference	2K–8K DZD

SUBSCRIPTION

13KDZD/mo

CLINIC

Mid-size clinic

Daily volume	100 pts
Monthly volume	3,000 pts
Storage / month	15 GB
AI inference	7K–25K DZD

SUBSCRIPTION

32KDZD/mo

HOSPITAL

CHU / large institution

Daily volume	200 pts
Monthly volume	6,000 pts
Storage / month	30 GB
AI inference	14K–55K DZD

SUBSCRIPTION

64KDZD/mo

OPTIMISATION

Response caching

40–60% hit rate on common clinical patterns.

OPTIMISATION

Single-call reports

Consolidated PDF generation cuts ~75% of per-report cost.

OPTIMISATION

Tiered model selection

Lightweight models for routine queries; large models for diagnoses.

OPTIMISATION

Query reduction

Drop default queries 4 → 2 per patient for straightforward cases.

The only multimodal AI for pediatric pneumonia in Algeria.

CRITERION	PNEUMO-AI	INTERNATIONAL EMR	TELEMEDICINE APPS	TRADITIONAL PRACTICE
AI diagnostic engine	Dual AI (GBM + DenseNet121)	None	Basic triage only	None
RAG clinical chatbot	Hybrid FAISS + BM25 + reranker	None	None	None
Algerian context	Built for Algeria (DZD, DZ law)	Foreign pricing	Partial	Yes
X-ray Grad-CAM	Visual explainability	None	None	None
PDF report generation	Automated, timestamped	Manual / generic	None	None
Data sovereignty	iBox (Algérie Télécom)	Foreign cloud	Foreign cloud	Local
Pricing model	13K–64K DZD / month	USD pricing (prohibitive)	Variable	N / A
LLM medical roadmap	Year 3 Algerian Medical LLM	None locally	None	None

MOAT 01

Local-first: DZD pricing, iBox hosting, Arabised UI.

MOAT 02

Real Algerian pediatric data — unavailable to foreign competitors.

MOAT 03

CNN + GBM + RAG in one workflow — no equivalent in market.

MOAT 04

File-based, locally hosted — aligns with Algerian health data law.

Three phases. Free first, then institutional.

Build credibility through clinical adoption — convert via institutional partnerships — scale to research and LLM products.

PHASE 01 · MONTHS 0–6

Free pilot adoption

- 10 selected pediatric clinics receive free 6-month access
- 30 individual cabinets — premium free tier
- Daily clinical-data collection in DZ context
- Build credibility with Algerian medical community

Investment: 2–4 M DZD

PHASE 02 · MONTHS 6–18

Paid SaaS conversion

- Convert pilots to paid Médecin Essentiel / Hôpital Pro
- Direct sales to Algerian CHUs and clinic networks
- Strategic partnerships: SAP, Algerian Pediatric Society
- Scientific publication of clinical results

Target: 50 paid subscribers

PHASE 03 · MONTHS 18–36

Scale & data products

- Launch Données Recherche — sell datasets to research labs
- Develop the Algerian Medical LLM (LLM Médical Algérien)
- Maghreb expansion: Tunisia, Morocco
- International ISO 13485 / WHO certification

Target Y3: 50–150 M DZD revenue

Subscription tiers under the KHIAT Startup framework. framework.

Three SaaS tiers for clinical use, plus two premium offers — research data and the Algerian Medical LLM.

TIER 01 · SAAS

Médecin Essentiel

12K DZD / month

- Up to 30 patients / day
- Dual AI + clinical chatbot
- Standard PDF reports
- iBox storage included

Annual offer: 130K DZD (2 months free)

RECOMMENDED

TIER 02 · SAAS

Hôpital Pro

45K DZD / month

- Up to 150 patients / day
- 5 simultaneous physicians
- Multimodal RAG + custom reports
- Priority Algerian-data training

Annual offer: 480K DZD

TIER 03 · SAAS

CHU Enterprise

120K+ DZD / month

- Unlimited capacity
- Multi-departmental, multi-site
- Dedicated DZ-data fine-tuning
- 24 / 7 SLA + dedicated support

3-year contract: from 4 M DZD

PREMIUM 01

Données Recherche — research data licensing

Anonymised pediatric clinical datasets sold to research labs and pharma. Initial bundle: 500 K – 5 M DZD. Custom CHU/INSP partnerships negotiated separately.

PREMIUM 02

LLM Médical — Algerian medical LLM

Trained on PNEUMO-AI's Algerian clinical corpus. Subscription from 8 K DZD/month for clinics; enterprise integration up to 6 M DZD; per-query API access.

The people behind PNEUMO-AI.

Combined expertise in AI research, clinical pediatrics, and business strategy.

FOUNDER & SUPERVISOR

Dr. Abderrahmane Khiat

PhD Computer Science · AI Systems & Digital Health

- Project Lead, Fraunhofer Institute IAIS (Germany)
- IT Project Manager, Hyundai AutoEver (South Korea)
- Lecturer & Researcher, University of Saida (Algeria)
- Founder, KHIAT Startup Framework

MEDICAL ADVISOR

Dr. Aimer Mohammed D.E.

Pediatric Clinical Validation

- Clinical validation of AI outputs
- Confirms diagnostic threshold safety
- Pediatric medicine & safety standards
- Liaison with Algerian medical community

PM & ML LEAD

Kassouar Fatima

CSV preprocessing pipeline
7-classifier training stack
Late-fusion manger.

DL ENGINEER

Labani Nabila N.E.

CNN training (DenseNet, ResNet, VGG)
Threshold optimisation
Grad-CAM visualisation
External validation

DATA ENGINEER

Bouhmidi Amina M.

X-ray preprocessing
Stratified split design
Augmentation pipeline
Cross-modal alignment

BUSINESS ANALYST

Miloudi Meroua A.

Market analysis (DZ healthcare)
Pricing & subscription strategy
Partnership outreach
Go-to-market planning

CONTACT & PARTNERSHIPS

Let's build pediatric AI for Algeria — together.

FOUNDER & SUPERVISOR

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University of Saida, Algeria

PROJECT REPOSITORY

Open-source on GitHub

https://github.com/AminaMar/pediatric-pneumonia-detection/tree/main

Notebooks · pipelines · documentation

LIVE PROTOTYPE

Web demo on request

Available for clinical evaluation

Grad-CAM · 3-tier output · PDF reports

OUR ASK

A pilot partnership with one Algerian pediatric hospital · Ethics committee approval to expand the clinical dataset · Access to ANDS / WHO EMRO grant programs · Introductions to clinical advisors and Algerian medical societies.